

Jai Sai Charan Kumar

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SUMMARY

Data Science aspirant skilled in ML, Deep Learning, NLP, and LLM-based systems. Experienced in RAG pipelines, transformer implementation, predictive modeling, and building Python- and Streamlit-based applications. Strong foundation in data preprocessing, model evaluation, SQL, and visualization. Focused on developing scalable, real-world AI solutions.

WORK EXPERIENCE

DataScienceIntern

CognifyzTechnologies—Remote

May 2025 – June 2025

- Analyzed the **Zomato Restaurant Dataset** to identify trends in customer ratings, pricing, and cuisine popularity using Python (Pandas, Matplotlib, Seaborn).
- Built and evaluated **predictive models** to forecast restaurant ratings and improve recommendation accuracy.
- Presented key insights and visualizations to stakeholders, supporting data-driven decisions.

PROJECTS

Voice Emotion Recognition

- Built a model to detect emotions (happy, sad, angry, etc.) from voice recordings.
- Extracted features like **MFCC** and chroma from audio and used models like **SVM** and **Random Forest**.
- Achieved over **80% accuracy** and created a basic interface for real-time predictions.

Sports Person Classifier

- Created a web app to identify sports personalities from images.
- Used **OpenCV** to detect faces, applied wavelet transform for features, and classified with an **SVM** model.
- Developed the backend with **Flask** and designed a simple frontend for image uploads and results.

RAG-Powered Webpage Ingestion & Q/A System

- Developed a full **RAG** pipeline to ingest URLs and convert raw HTML into structured, searchable content.
- Built title-aware chunking and metadata extraction using **BeautifulSoup** and Unstructured for precise document segmentation.
- Created dense embeddings with **Google Generative AI** and indexed them in **ChromaDB** for fast semantic retrieval.
- Implemented a **Streamlit** UI enabling URL ingestion, retrieval, and Gemini-based, source-grounded question answering.

Custom LLM Built From Scratch (GPT-2 Architecture)

- Implemented a full GPT-2-style **transformer (attention, FFN, GELU, LayerNorm, causal masking)** entirely from scratch
- Successfully loaded and remapped **256M pretrained weights** for inference compatibility.
- Built a **Streamlit**-based chatbot interface enabling instruction-style prompting.
- Implemented nucleus sampling (top-p) for controlled text generation.

SKILLS

Languages: Python, SQL

Python Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, NLTK, Matplotlib, Seaborn, Hugging Face, langchain

Technologies: Machine Learning, Deep Learning, Basics of NLP, LLM, RAG

Tools & Concepts: Model Evaluation & Validation, Data Visualization, Statistical Analysis, Data Mining

Languages Known: English, Telugu, Hindi

EDUCATION

Bachelor Degree in CSE

Rajiv Gandhi University of Knowledge Technologies - Ongole, India - 2022 - 2026 - 9.2 / 10.0

Pre-University Education

Rajiv Gandhi University of Knowledge Technologies - Ongole, India - 2020 - 2022 - 9.89 / 10.0

CERTIFICATIONS

Introduction to Deep Learning - Certification course by NPTEL

Object Oriented System Design - Certification course by NPTEL

Python for Data Science, AI & Development - IBM by Coursera

Responsible and safe AI - Certification course by NPTEL

HOBBIES

Weightlifting - Passionate about strength training and maintaining physical fitness through consistent gym workouts.

Tech Exploration - Enjoy exploring emerging technologies, tools, and innovations in the field of computer science and AI.